SOLUTIONS: STABILITY AND CONTROL OF NONLINEAR SYSTEMS MASTER IN CONTROL

1. Exercise

a) The vector field $f(x_1, x_2)$ is a polynomial with respect to both variables. Hence it is a locally Lipschitz function and by Caratheodory's theorem solutions exist and are unique (over their maximal time-interval of definition). [2]

b) The nullclines of the vector-field can be computed as shown below:

$$\mathcal{N}_1 = \{x : x_2 = x_1^3 - x_1/4\}.$$

This is a cubic which intersects the x_1 axis in $\pm 1/2$ and 0. All solutions above the cubic move eastwards and those below west. For the second nullcline we see:

$$\mathcal{N}_2 = \{x : x_1^2 + x_2^2 = 1/4\}.$$

The second nullcline is a circle of radius $1/2$, centered at 0. All solutions inside the circle move south and those outside move north. [2]

c) There are 4 regions delimited by the nullclines. The direction of the vector-field in each one of them is shown in Fig. 1.1. [2]

d) From the direction of the arrows we notice that all regions have arrows pointing outwards; hence none of them is forward invariant. [2]

e) Nullclines intersect in 2 points, $[1/2, 0]'$ and $[-1/2, 0]'$ which are the only equilibria of the system. [2]

f) To linearize the dynamics we first compute the Jacobian matrix $\frac{\partial f}{\partial x}$ at an arbitary value of x.

$$\frac{\partial f}{\partial x}(x) = \begin{bmatrix} -12x_1^2 + 1 & 4 \\ 8x_1 & 8x_2 \end{bmatrix}.$$

At the equilibrium $x_{e1} = [1/2, 0]'$ we see that:

$$\frac{\partial f}{\partial x}(x_{e1}) = \begin{bmatrix} -2 & 4 \\ 4 & 0 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_1(s) = s^2 + 2s - 16$, whose roots are $-1 \pm \sqrt{17}$. Two real eigenvalues of opposite sign correspond to the phase portrait of a saddle equilibrium (hyperbolic). At the equilibrium $x_{e2} = [-1/2, 0]'$ we have the Jacobian:

$$\frac{\partial f}{\partial x}(x_{e2}) = \begin{bmatrix} -2 & 4 \\ -4 & 0 \end{bmatrix}.$$

Its characteristic polynomial is $\chi_2(s) = s^2 + 2s + 16$ with roots in $-1 \pm \sqrt{15}j$. Two complex conjugates eigenvalues with negative real part correspond to the phase portrait of a stable focus (hyperbolic). [3]

g) The set R is sketched in the Fig. 1.2. Notice that since all arrows at his boundary point towards its interior, the set is forward invariant. All solutions starting inside R stay in R forever. In particular both $x_1(t)$ and $x_2(t)$ are monotonically increasing. Hence their limit exist. Consider variable x_2. If its limit is finite,

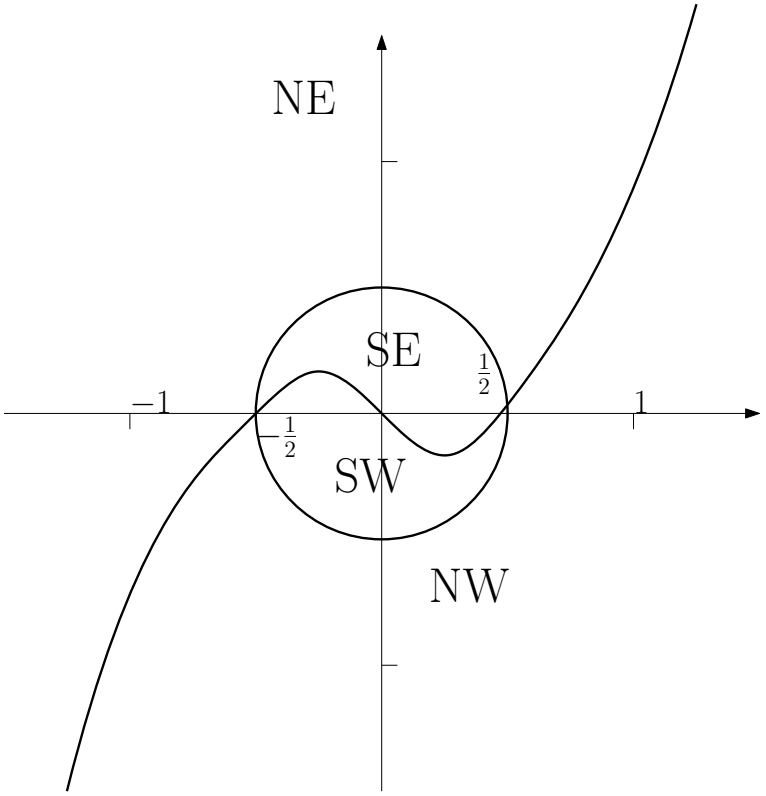


Figure 1.1 Nullclines and vector-field orientations

say $\bar{x_2}$ we remark that the set $R \cap \{x : x_2 \leq \bar{x}_2\}$ is bounded and therefore also $x_1(t)$ would be uniformly bounded. In particular, then it would admit a finite limit $\bar{x}_1$. This, however, means that $[\bar{x}_1, \bar{x}_2]$ is an equilibrium point, which is a contradiction, since there are only 2 equilibria, and none in R. Hence, we can conclude that $x_2(t)$ diverges to ∞ and the ω limit set of any solutions initiated in R is empty. [4]

h) The global phase portrait is sketched in Fig. 1.3. [2]

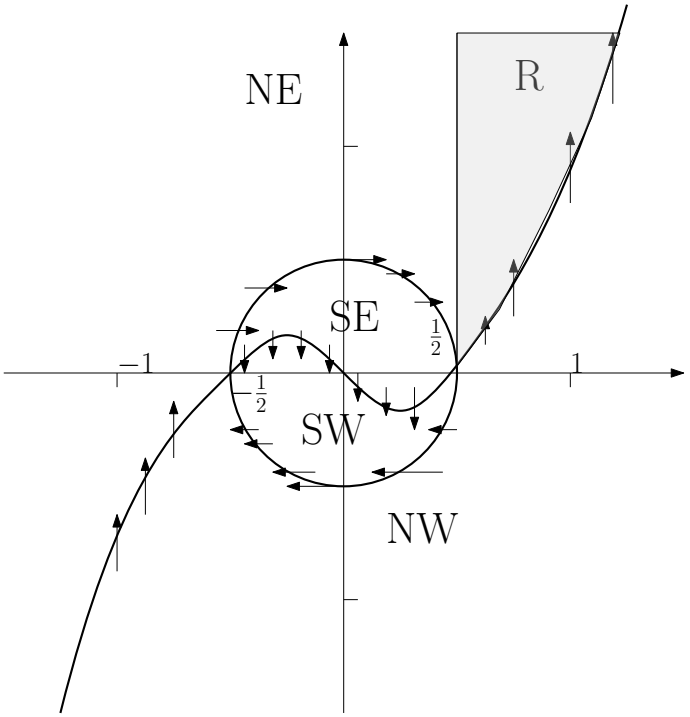


Figure 1.2 Region *R* and vector field orienation on Nullclines

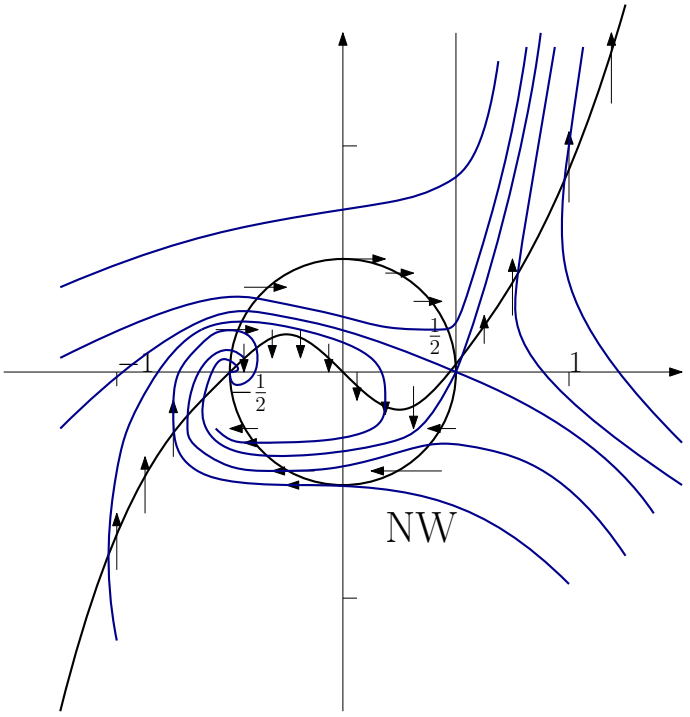


Figure 1.3 Global phase portrait, a sketch

2. Exercise

a) We choose the state variable x as the output and its derivative, $x(t) = [y(t), \dot{y}(t)]' = [x_1(t), x_2(t)]'$. Accordingly we write state-space equations as:

$$\begin{aligned} \dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= -\sin^2(x_2(t))x_2(t) - x_1^3(t) + u(t). \end{aligned}$$

[3]

b) For each input value u, the corresponding equilibrium(s) are found as solutions of the equation:

$$\begin{aligned} 0 &= x_2 \\ 0 &= -\sin^2(x_2)x_2 - x_1^3 + u. \end{aligned}$$

From the first equation we see that $x_2 = 0$, and substituting this value in the next equation we obtain $x_1^3 = u$, viz. there exists a unique equilibrium:

$$x_e(u) = \begin{bmatrix} \sqrt[3]{u} \\ 0 \end{bmatrix}.$$

[2]

c) The kinetic energy of a unit mass is $\dot{y}^2/2$, while the potential energy of a cubic spring is given by $y^4/4$ (viz. the integral of y^3 between 0 and y). Accordingly we may choose $V(x)$ as:

$$V(x) = \frac{x_2^2}{2} + \frac{x_1^4}{4}.$$

[2]

d) $V(x)$ is clearly positive semi-definite, as the sum of two non-negative terms (being monomials of even powers of x_i). Moreover:

$$x \neq 0 \Rightarrow x_1 \neq 0 \text{ or } x_2 \neq 0.$$

In the first case we have $x_1^4 > 0$, and therefore $V(x) > 0$, while in the second case we see $x_2^2 > 0$ and therefore the same conclusion $V(x) > 0$ holds. This proves positive-definiteness of V.
To show radial unboundedness, for an arbitrary $M > 0$ we consider the set:

$$S_M = \{x : V(x) \leq M\}.$$

We want to show that this is a bounded set. Any point in S_M must fulfill:

$$x_2^2/2 \leq M \text{ and } x_1^4/4 \leq M.$$

From the first inequality we get a bound on x_2, in particular, $|x_2| \leq \sqrt{2M}$. From the second one we see that $|x_1| \leq \sqrt[4]{4M}$. Overall both x_1 and x_2 are bounded as a function of M and therefore the set S_M is bounded. [3]

e) Computation of $\dot{V}(x) = \frac{\partial V}{\partial x}(x)f(x)$ yields:

$$\begin{aligned} \dot{V}(x) = x_1^3\dot{x}_1 + x_2\dot{x}_2 &= x_1^3x_2 - x_1^3x_2 - x_2^2\sin^2(x_2) \\ &= -x_2^2\sin^2(x_2) \leq 0. \end{aligned}$$

Hence $\dot{V}(x)$ is negative semi-definite. Moreover its Kernel is given by $\{x : x_2 = k\pi, k \in \mathbb{Z}\}$. Hence $\dot{V}$ is not negative definite. [2]

f) We would like to show that 0 is globally asymptotically stable through the use of Lasalle's invariance criterion. To this end we express its kernel as:

$$\text{Ker}[\dot{V}] = \{x : x_2 - k\pi = 0\} := K_0.$$

Any solution which belongs to K_0 for all t must also belong to:

$$K_1 = \{x : x_2 = k\pi \text{ and } -\sin^2(x_2)x_2 - x_1^3 = 0\}$$

$$= \{x : x_2 = k\pi \text{ and } x_1 = 0\}.$$

Similarly, any solution belonging to K_1 for all t must also belong to:

$$K_2 = \{x : x_2 = k\pi, x_1 = 0, 0 = \dot{x}_1 = x_2\} = \{0\}.$$

Hence the largest invariant set contained in $\text{Ker}[\dot{V}(x)]$ is the origin. By Lasalle's invariance criterion 0 is a globally asymptotically stable equilibrium. [3]

g) We define shifted coordinates according to:

$$\tilde{x}_1 = x_1 - \sqrt[3]{u}, \qquad \tilde{x}_2 = x_2$$

Taking derivatives with respect to time yields:

$$\begin{aligned} \dot{\tilde{x}}_1 &= \dot{x}_1 = x_2 = \tilde{x}_2 \\ \dot{\tilde{x}}_2 &= \dot{x}_2 = -\sin^2(x_2)x_2 - x_1^3 = -\sin^2(\tilde{x}_2)\tilde{x}_2 - (\tilde{x}_1 + \sqrt[3]{u})^3 + u. \end{aligned}$$

[2]

h) The system in $\tilde{x}$ coordinates has a similar physical interpretation, however the expression of the elastic force is changed and is slightly more complicated. This alters the expression for the potential energy. In particular, the elastic force is given as:

$$(\tilde{x}_1 + \sqrt[3]{u})^3 - u = \tilde{x}_1^3 + 3\tilde{x}_1^2\sqrt[3]{u} + 3\tilde{x}_1\sqrt[3]{u^2}.$$

The new potential energy reads:

$$\int_0^{\tilde{x}_1} [\tilde{x}_1^3 + 3\tilde{x}_1^2\sqrt[3]{u} + 3\tilde{x}_1\sqrt[3]{u^2}]\, d\tilde{x}_1 = \frac{1}{4}\tilde{x}_1^4 + \sqrt[3]{u}\,\tilde{x}_1^3 + \frac{3}{2}\sqrt[3]{u^2}\,\tilde{x}_1^2.$$

Hence, the Lyapunov function in $\tilde{x}$ coordinates reads:

$$V(\tilde{x}) = \frac{1}{2}\tilde{x}_2^2 + \frac{1}{4}\tilde{x}_1^4 + \sqrt[3]{u}\,\tilde{x}_1^3 + \frac{3}{2}\sqrt[3]{u^2}\,\tilde{x}_1^2.$$

[3]

3. Exercise

a) We let x_1 be the state of system Σ_1 and interpret x_2 in the equation for $\dot{x}_1$ as the input signal. As a result the equation for Σ_1 reads:

$$\dot{x}_1 = -x_1 + d_1^3$$

Similarly, we let x_2 be the state of system Σ_2 and interpret x_1 in the equation for $\dot{x}_2$ as the input signal. As a result the equation for Σ_2 reads:

$$\dot{x}_2 = -x_2^9 + ad_2^3 + d$$

[4]

b) To show ISS of Σ_1 we pick the Lyapunov function $V_1(x_1) = x_1^2/2$. Clearly $V_1(x_1)$ is positive definite and radially unbounded with:

$$\frac{|x_1|^2}{2} \leq V_1(x_1) \leq \frac{|x_1|^2}{2}.$$

Computing its derivative along solutions of the Σ_1 equation yields:

$$\dot{V}_1 = x_1\dot{x}_1 = -x_1^2 + x_1 d_1^3 \leq -|x_1|^2 + |x_1||d_1|^3$$
$$= -\varepsilon|x_1|^2 - (1-\varepsilon)|x_1|^2 + |x_1||d_1|^3.$$

where $\varepsilon > 0$ is a small positive constant to be assigned later. Clearly, the following implication holds:

$$(1-\varepsilon)|x_1|^2 \geq |x_1||d_1|^3 \Rightarrow \dot{V}_1 \leq -\varepsilon|x_1|^2.$$

In particular then, working on the first inequality we can show that:

$$|x_1| \geq \frac{|d_1|^3}{1-\varepsilon} \Rightarrow \dot{V}_1 \leq -\varepsilon|x_1|^2.$$

This proves that V_1 is an ISS-Lyapunov function and that $\gamma_1(r) = r^3/(1-\varepsilon)$ is a suitable $\mathscr{K}_\infty$ gain function for Σ_1. [4]

c) To show ISS of Σ_2 we pick the Lyapunov function $V_2(x_2) = x_2^2/2$. Clearly $V_2(x_2)$ is positive definite and radially unbounded with:

$$\frac{|x_2|^2}{2} \leq V_2(x_2) \leq \frac{|x_2|^2}{2}.$$

Computing its derivative along solutions of the Σ_2 equation yields:

$$\dot{V}_2 = x_2\dot{x}_2 = -x_2^{10} + ax_2 d_2^3 + ax_2 d$$

In order to compute the gain from d_2 to x_2 we let $d = 0$, so that

$$\dot{V}_2 \leq -|x_2|^{10} + a|x_2||d_2|^3$$
$$= -\varepsilon|x_2|^{10} - (1-\varepsilon)|x_2|^{10} + |a||x_2||d_2|^3.$$

The following implication holds:

$$(1-\varepsilon)|x_2|^{10} \geq |a||x_2||d_2|^3 \Rightarrow \dot{V}_2 \leq -\varepsilon|x_2|^{10}.$$

In particular then working on the first inequality we can show that:

$$|x_2| \geq \sqrt[3]{|d_2|}\sqrt[9]{\frac{|a|}{1-\varepsilon}} \Rightarrow \dot{V}_2 \leq -\varepsilon|x_2|^{10}.$$

This proves that V_2 is an ISS-Lyapunov function and that $\gamma_2(r) = \sqrt[3]{r}\sqrt[9]{|a|/(1-\varepsilon)}$ is a suitable $\mathscr{K}_\infty$ gain function for Σ_2. [4]

d) To verify the small gain condition we compute $\gamma_1(\gamma_2(r))$:

$$\gamma_1(\gamma_2(r)) = \gamma_1(\sqrt[3]{r}\sqrt[9]{|a|/(1-\varepsilon)}) = \left(\sqrt[3]{r}\sqrt[9]{|a|/(1-\varepsilon)}\right)^3/(1-\varepsilon)$$

$$= r\sqrt[9]{|a|/(1-\varepsilon)}/(1-\varepsilon)$$

The condition $\gamma_1(\gamma_2(r)) < r$ for all $r > 0$ is fulfilled provided $|a| < 1$, (and provided ε is sufficiently close to 0). Hence, by the small-gain theorem the system Σ is ISS with respect to the exogenous input d. [4]

e) To verify lack of asymptotic stability of the origin for $a = 1$ (and hence tightness of the small gain theorem) we compute the set of equilibria for $d = 0$. The equations read:

$$\begin{cases} 0 &= -x_1 + x_2^3 \\ 0 &= -x_2^9 + x_1^3 \end{cases}$$

Notice that whenever the first equation is fulfilled the second also holds. Hence we have a curve of equilibria; in particular every point of type $[c^3, c]$ with $c \in \mathbb{R}$ is an equilibrium. For this reason, the origin is not locally asymptotically stable. In fact there exist initial conditions, arbitrarily close to it, whose corresponding solutions are not attracted to the origin. [4]

4. Exercise

a) Consider the output $y = x_1$. To compute its relative degrees we start differentiating it along solutions of the nonlinear system:

$$\dot{y} = \dot{x}_1 = x_3 + \sin(x_2)$$

$$\ddot{y} = \dot{x}_3 + \cos(x_2)\dot{x}_2 = \sin(x_3) + (2 + \cos(x_2))u + \cos(x_2)(-x_2^3 + x_1^3).$$

Notice that $\ddot{y}$ is a function of u. Hence the relative degree is 2. [2]

Moreover the coefficient of u is $(2 + \cos(x_2)) > 0$ for all $x \in \mathbb{R}^3$. Hence the relative degree is globally defined. [1]

b) Consider the state-feedback control:

$$u = -\frac{\sin(x_3) + \cos(x_2)(-x_2^3 + x_1^3) + v}{2 + \cos(x_2)},$$

where v is an auxiliary control input. Substituting in the expression for $\ddot{y}$ we get that:

$$\ddot{y} = v,$$

hence we have now a linear input-output relationship between v and y. [3]

c) To stabilize the equation of a double integrator it is enough to use a PD controller. In particular:

$$v = -k_p y - k_d \dot{y} = -k_p x_1 - k_d(x_3 + \sin(x_2))$$

would stabilize the system at $y = 0$, $\dot{y} = 0$ for any choice of gains $k_p > 0$ and $k_d > 0$. [3]

d) To highlight the normal form we choose the following set of coordinates: $z = [y, \dot{y}]'$ and $\xi = x_2$. In particular then, letting $\tilde{x} = [x_1, x_3 + \sin(x_2), x_2]'$ we see that:

$$\begin{aligned} \dot{z}_1 &= z_2 \\ \dot{z}_2 &= v \\ \dot{\xi} &= -\xi^3 + z_1^3. \end{aligned}$$

Notice that the internal dynamics corresponding to $\dot{\xi}$ are input independent, as it should be in the normal form.

e) If we regard the internal dynamics as a scalar system with state ξ and input z_1 we see that this is an ISS system. In particular:

$$\dot{\xi} = -\xi^3 + d_1^3,$$

where d_1 has replaced z_1 to emphasize that we now regard the signal as an input disturbance. [2]

To study ISS of this system it is enough to take $V(\xi) = \xi^2/2$ and show that this is an ISS Lyapunov function. Its derivative along solutions reads:

$$\dot{V} = \xi\dot{\xi} = -\xi^4 + \xi d_1^3 \leq -|\xi|^4 + |\xi||d_1|^3$$

$$\leq -\frac{1}{2}|\xi|^4 - \frac{1}{2}|\xi|^4 + |\xi||d_1|^3$$

In particular then:

$$|\xi|^3 \geq 2|d_1|^3 \Rightarrow \dot{V} \leq -\frac{1}{2}|\xi|^4.$$

Finally,

$$|\xi| \geq \sqrt[3]{2}|d_1| \Rightarrow \dot{V} \leq -\frac{1}{2}|\xi|^4,$$

which proves that V is an ISS Lyapunov function and the internal dynamics are ISS with respect to input disturbance d_1. [2]

Hence, any linear feedback stabilizing the y subsystem, will result in a cascade of two ISS systems, which is still ISS and therefore globally asymptotically stable. [2]